```sql
/*
    RaceDay Database Script
    PROG6212 Part 1 - Section C
    SQL Server / SSMS
*/

-- Create the database
IF DB_ID('RaceDayDB') IS NOT NULL
BEGIN
    ALTER DATABASE RaceDayDB SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE RaceDayDB;
END;
GO

CREATE DATABASE RaceDayDB;
GO

USE RaceDayDB;
GO

/* ========================
   1. USER
   ======================== */
CREATE TABLE [User]
(
    UserID INT IDENTITY(1,1) NOT NULL,
    FirstName NVARCHAR(50) NOT NULL,
    LastName NVARCHAR(50) NOT NULL,
    Email NVARCHAR(100) NOT NULL,
    PasswordHash NVARCHAR(255) NOT NULL,
    Role NVARCHAR(20) NOT NULL,
    PhoneNumber NVARCHAR(20) NULL,

    CONSTRAINT PK_User PRIMARY KEY (UserID),
    CONSTRAINT UQ_User_Email UNIQUE (Email),
    CONSTRAINT CK_User_Role CHECK (Role IN ('Organiser',
      'Participant'))
);
GO

/* ========================
   2. PARTICIPANT PROFILE
   ======================== */
CREATE TABLE ParticipantProfile
(
    ProfileID INT IDENTITY(1,1) NOT NULL,
    UserID INT NOT NULL,
    DateOfBirth DATE NULL,
    ProfilePictureURL NVARCHAR(500) NULL,

    CONSTRAINT PK_ParticipantProfile PRIMARY KEY (ProfileID),
    CONSTRAINT UQ_ParticipantProfile_User UNIQUE (UserID),
    CONSTRAINT FK_ParticipantProfile_User
```

```
53          FOREIGN KEY (UserID) REFERENCES [User](UserID)
54  );
55  GO
56
57  /* =========================
58     3. EVENT TYPE
59     ========================= */
60  CREATE TABLE EventType
61  (
62      EventTypeID INT IDENTITY(1,1) NOT NULL,
63      TypeName NVARCHAR(20) NOT NULL,
64
65      CONSTRAINT PK_EventType PRIMARY KEY (EventTypeID),
66      CONSTRAINT UQ_EventType_TypeName UNIQUE (TypeName),
67      CONSTRAINT CK_EventType_TypeName CHECK (TypeName IN ('Run', '
          Walk', 'Cycle'))
68  );
69  GO
70
71  /* =========================
72     4. EVENT
73     ========================= */
74  CREATE TABLE Event
75  (
76      EventID INT IDENTITY(1,1) NOT NULL,
77      OrganiserID INT NOT NULL,
78      EventTypeID INT NOT NULL,
79      Name NVARCHAR(150) NOT NULL,
80      Description NVARCHAR(500) NULL,
81      EventDate DATE NOT NULL,
82      Location NVARCHAR(200) NOT NULL,
83      Distance DECIMAL(6,2) NOT NULL,
84      BannerImageURL NVARCHAR(500) NULL,
85
86      CONSTRAINT PK_Event PRIMARY KEY (EventID),
87      CONSTRAINT CK_Event_Distance CHECK (Distance > 0),
88      CONSTRAINT FK_Event_Organiser
89          FOREIGN KEY (OrganiserID) REFERENCES [User](UserID),
90      CONSTRAINT FK_Event_EventType
91          FOREIGN KEY (EventTypeID) REFERENCES EventType(EventTypeID)
92  );
93  GO
94
95  /* =========================
96     5. CATEGORY
97     ========================= */
98  CREATE TABLE Category
99  (
100     CategoryID INT IDENTITY(1,1) NOT NULL,
101     EventID INT NOT NULL,
102     CategoryName NVARCHAR(100) NOT NULL,
103     MinimumAge INT NULL,
104     MaximumAge INT NULL,
```

```
105     Distance DECIMAL(6,2) NULL,
106
107     CONSTRAINT PK_Category PRIMARY KEY (CategoryID),
108     CONSTRAINT CK_Category_AgeRange CHECK
109     (
110         (MinimumAge IS NULL AND MaximumAge IS NULL)
111         OR
112         (MinimumAge IS NOT NULL AND MaximumAge IS NOT NULL
113          AND MinimumAge >= 0 AND MaximumAge >= MinimumAge)
114     ),
115     CONSTRAINT CK_Category_Distance CHECK (Distance IS NULL OR
          Distance > 0),
116     CONSTRAINT FK_Category_Event
117         FOREIGN KEY (EventID) REFERENCES Event(EventID)
118 );
119 GO
120
121 /* =========================
122    6. ENROLMENT
123    ========================= */
124 CREATE TABLE Enrolment
125 (
126     EnrolmentID INT IDENTITY(1,1) NOT NULL,
127     ParticipantID INT NOT NULL,
128     EventID INT NOT NULL,
129     CategoryID INT NOT NULL,
130     EnrolmentDate DATETIME2 NOT NULL CONSTRAINT DF_Enrolment_Date
          DEFAULT SYSDATETIME(),
131     Status NVARCHAR(20) NOT NULL CONSTRAINT DF_Enrolment_Status
          DEFAULT 'Pending',
132
133     CONSTRAINT PK_Enrolment PRIMARY KEY (EnrolmentID),
134     CONSTRAINT UQ_Enrolment_Participant_Event UNIQUE (ParticipantID,
          EventID),
135     CONSTRAINT CK_Enrolment_Status CHECK (Status IN ('Pending',
          'Confirmed', 'Cancelled')),
136     CONSTRAINT FK_Enrolment_Participant
137         FOREIGN KEY (ParticipantID) REFERENCES [User](UserID),
138     CONSTRAINT FK_Enrolment_Event
139         FOREIGN KEY (EventID) REFERENCES Event(EventID),
140     CONSTRAINT FK_Enrolment_Category
141         FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)
142 );
143 GO
144
145 /* =========================
146    7. RESULT
147    ========================= */
148 CREATE TABLE Result
149 (
150     ResultID INT IDENTITY(1,1) NOT NULL,
151     EnrolmentID INT NOT NULL,
152     FinishTime TIME NOT NULL,
```

```
153     FinishPosition INT NOT NULL,
154     ResultDate DATETIME2 NOT NULL CONSTRAINT DF_Result_Date DEFAULT
          SYSDATETIME(),
155
156     CONSTRAINT PK_Result PRIMARY KEY (ResultID),
157     CONSTRAINT UQ_Result_Enrolment UNIQUE (EnrolmentID),
158     CONSTRAINT CK_Result_Position CHECK (FinishPosition > 0),
159     CONSTRAINT FK_Result_Enrolment
160         FOREIGN KEY (EnrolmentID) REFERENCES Enrolment(EnrolmentID)
161 );
162 GO
163
164 /* ========================
165    SAMPLE DATA
166    Minimum required:
167    - 2 Organisers
168    - 2 Participants
169    - 3 Events
170    - Categories for each event
171    - Sample enrolments
172    ======================== */
173
174 -- Users: 2 Organisers + 2 Participants
175 INSERT INTO [User]
176     (FirstName, LastName, Email, PasswordHash, Role, PhoneNumber)
177 VALUES
178     ('Thabo', 'Mokoena', 'thabo.mokoena@raceday.co.za',
          'HASHED_PASSWORD_001', 'Organiser', '0825551001'),
179     ('Lerato', 'Naidoo', 'lerato.naidoo@raceday.co.za',
          'HASHED_PASSWORD_002', 'Organiser', '0825551002'),
180     ('Sipho', 'Dlamini', 'sipho.dlamini@example.com',
          'HASHED_PASSWORD_003', 'Participant', '0825552001'),
181     ('Amahle', 'Khumalo', 'amahle.khumalo@example.com',
          'HASHED_PASSWORD_004', 'Participant', '0825552002');
182 GO
183
184 -- Participant profiles
185 INSERT INTO ParticipantProfile
186     (UserID, DateOfBirth, ProfilePictureURL)
187 VALUES
188     (3, '1998-04-15', NULL),
189     (4, '2002-09-21', NULL);
190 GO
191
192 -- Event types
193 INSERT INTO EventType (TypeName)
194 VALUES
195     ('Run'),
196     ('Walk'),
197     ('Cycle');
198 GO
199
200 -- Events
```

```
INSERT INTO Event
    (OrganiserID, EventTypeID, Name, Description, EventDate, Location, Distance, BannerImageURL)
VALUES
    (1, 1, 'Pretoria City Run', 'A community road running event through Pretoria.', '2026-10-10', 'Pretoria, Gauteng', 10.00, NULL),
    (1, 2, 'Soweto Community Walk', 'A family-friendly community walking event.', '2026-10-24', 'Soweto, Gauteng', 5.00, NULL),
    (2, 3, 'Joburg Cycle Challenge', 'A road cycling event for recreational and competitive cyclists.', '2026-11-07', 'Johannesburg, Gauteng', 40.00, NULL);
GO

-- Categories for each event
INSERT INTO Category
    (EventID, CategoryName, MinimumAge, MaximumAge, Distance)
VALUES
    (1, 'Under 20', 13, 19, NULL),
    (1, 'Senior', 20, 39, NULL),
    (1, '40 Plus', 40, 99, NULL),
    (2, 'Family Walk', NULL, NULL, 5.00),
    (2, 'Senior Walk', 60, 99, 5.00),
    (3, 'Open Cycle', 18, 59, 40.00),
    (3, 'Masters Cycle', 60, 99, 40.00);
GO

-- Sample enrolments
INSERT INTO Enrolment
    (ParticipantID, EventID, CategoryID, EnrolmentDate, Status)
VALUES
    (3, 1, 2, '2026-08-20 09:15:00', 'Confirmed'),
    (4, 1, 2, '2026-08-21 10:30:00', 'Confirmed'),
    (3, 2, 4, '2026-08-22 11:00:00', 'Confirmed'),
    (4, 2, 4, '2026-08-23 14:20:00', 'Pending'),
    (3, 3, 6, '2026-08-24 08:45:00', 'Confirmed');
GO

-- No sample results inserted because the seeded events are upcoming.
/* =========================
   VERIFICATION QUERIES
   ========================= */
SELECT * FROM [User];
SELECT * FROM ParticipantProfile;
SELECT * FROM EventType;
SELECT * FROM Event;
SELECT * FROM Category;
SELECT * FROM Enrolment;
SELECT * FROM Result;
GO
```